Adrián Pabón Mendoza

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Target Role

Senior AI Engineer (LLMs, RAG & AI Agents) | **AWS Certified Solutions Architect – Associate** | Electronics Engineer (Cum Laude) | Python, Java, FastAPI, Docker, Kubernetes, Terraform, AWS/Azure/OCI

Professional Summary

Senior Data and AI Engineer shipping LLM, RAG, and agentic systems to production since 2024. AWS Certified Solutions Architect – Associate. I build agentic architectures and RAG pipelines on vector databases (pgvector, Pinecone, OpenSearch), and evaluate LLM applications with custom datasets and LLM-as-a-judge. Scaled NAIA from a 500-user pilot to 13,000+ users with OpenAI LLMs, Redis, and Microsoft Graph/Azure. Built an agentic pipeline that migrates stored procedures to Java microservices, cutting migration documentation time from **2 days to 2 hours**. I deploy on AWS with serverless, event-driven architectures, Docker, Kubernetes, and Terraform.

Skills

LLMs & Agents: OpenAI, Amazon Bedrock, LangChain, LangGraph, Hugging Face Transformers, multi-agent orchestration, tool-calling, routing, prompt engineering.
RAG & Vector Search: embeddings, chunking, retrieval, reranking; pgvector, Pinecone, ChromaDB, FAISS, OpenSearch, Bedrock Knowledge Bases.
LLM Evaluation: custom evaluation datasets, LLM-as-a-judge, hallucination and answer-relevance metrics, prompt iteration.
Backend & Migrations: Python, Java, FastAPI/Flask, REST APIs, Redis, SQL, hexagonal architecture, stored procedure to microservice migration.
Data: data pipelines, DynamoDB data modeling, PostgreSQL, MongoDB, Databricks.
Cloud & DevOps: AWS (Lambda, Step Functions, Bedrock, SageMaker, DynamoDB, EventBridge, API Gateway, Cognito, EC2/S3/RDS), Terraform (IaC), Docker, Kubernetes (pod scaling and deployment configuration), Git, Azure, Oracle Cloud, IBM Cloud.
ML: PyTorch, scikit-learn, fine-tuning.

Experience

nyxn — Remote, Colombia
Senior Data and AI Engineer — *Feb 2026 – Present*

- Built an agentic architecture that migrates core banking stored procedures to Java 25 microservices with hexagonal architecture: SP and dependency analysis, logic migration, and generation of adapters, models, and utilities.
- Cut technical documentation time for SP migrations (Sonar, Prisma, Veracode, unit tests) from **2 days to 2 hours**.
- Designed and orchestrated a serverless, event-driven AWS product to track open roles across the company (Lambda, DynamoDB, EventBridge, S3, Bedrock, API Gateway, Cognito, SES), with infrastructure as code in **Terraform**.
- Built the product's data pipelines and designed its DynamoDB data model.

Optimize IT SAS — Barranquilla, Colombia
AI Product Specialist / AI Engineer — *Jul 2024 – Feb 2026*

- Delivered an LLM/RAG API with embeddings + vector search, iterating on chunking strategy, prompt templates, and retrieval parameters to improve grounded responses in internal workflows.
- Improved theft detection in micro-market systems using Computer Vision, reducing manual camera review time by **80%**.
- Trained and evaluated ML models for production use cases (classification/regression), tracking key metrics **(F1, precision, recall, MAE)** and validating with hold-out sets.
- Automated database backup workflows on OCI (scheduling, validation, notifications) with serverless functions, improving reliability, traceability, and reducing manual intervention.

Universidad del Norte — Barranquilla, Colombia
AI Researcher and Developer — *Jan 2024 – Nov 2025*

- Reduced administrative response time from hours/days to **3–5 seconds** by deploying NAIA (13,000+ users), enabling self-service for routine queries.
- Designed a modular monolith integrating OpenAI LLMs, Redis state, and external services via Azure; implemented secure access through Microsoft Graph API (email/calendar/authentication).
- Implemented RAG pipelines (chunking, embeddings, retrieval, reranking when applicable) and defined evaluation sets to

reduce hallucinations and improve answer relevance.

Featured Projects

AI Agent Implementation for Medical Call Center *2026*

Intelligent voice agents to improve customer care experience at a clinical laboratory.

- Executed outbound campaigns for a medical call center.
- Implemented **Tool-Calling** logic for real-time appointment and lab result validation, connecting the agent directly to the medical center's databases.
- Impact: **50%** reduction in call center agents' time spent on routine calls, freeing them for higher-impact interactions.

NAIA — Nimble Artificial Intelligence Assistant *2024–2025*

Portfolio link | Multi-role assistant with LLM/RAG and integrations.

- Architecture: OpenAI LLMs + Redis + Azure services + Microsoft Graph API integration.
- Scale: 13,000+ users; definition of operational flows and monitoring for production use.
- Impact: conversational responses in **3–5 seconds** (web architecture baseline).

Theft Detection (Computer Vision) *2024*

Production CV pipeline for micro-market security.

- Data: dataset curation and labeling; model training/validation with CV metrics **(mAP/F1)** and hold-out evaluation.
- Impact: **80%** reduction in manual camera review time.

Publications & Research

- **NAIA: A Robust AI Framework for Academic Assistance** — *Systems* (2025).
- **Architectures for AI-Powered Virtual Assistants** — *IEEE TEMSCON LATAM* (2025).
- **NAIA: A Multi-Technology Virtual Assistant** — *IEEE Access* (2025).
- **Itinerary Assignment Optimization** — *IEEE COLCOM* (2024).

Education

Universidad del Norte Barranquilla, Colombia

B.Eng. in Electronics Engineering (Cum Laude) *2020 – 2025*

Certifications & Recognition

- **AWS Certified Solutions Architect – Associate** — Amazon Web Services (Oct 2026 – Oct 2029) | Verify on Credly
- **Databricks Fundamentals** (2025) | **NVIDIA: Applications of AI for Anomaly Detection** (2025) | **NVIDIA: Generative AI with Diffusion Models** (2025)
- **Speaker:** Cátedra Europa 2024 ("Computer Vision: A World of Possibilities"). | **Recognition:** Best Science Project (Colciencias, 2019).

Languages

Spanish (Native) | English (Professional / C1)